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GUIDANCE DOCUMENT 5c: SECTOR-SPECIFIC GUIDANCE DOCUMENT ON FERTILISERS

Disclaimer: This guidance document is not legally binding and it is of an explanatory nature. Legal provisions of Union law take precedence over the contents of this document and should always be consulted. The authentic texts of the EU legal instruments are those published in the Official Journal of the European Union. This guidance was prepared by Commission services and does not represent an official position of the Commission. Only the Court of Justice of the European Union is competent to authoritatively interpret Union law.

VERSION HISTORY

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1 PREFACE

1.1 Structure and content of the guidance documents

This series of documents has been written to support stakeholders by explaining the requirements of the CBAM Regulation in a non-legislative language. This series of guidance documents should benefit a variety of stakeholders. A specific focus has been put on clarifying the **requirements for operators of installations producing CBAM goods outside of the EU for the definitive period**.

- **Guidance 1** provides an overview of the main changes to the CBAM methodology between the definitive period and the concluded transitional period. It also introduces the CBAM and an overview of the compliance cycle, roles and responsibilities, as well as milestones and deadlines for operators of installations outside the EU during the definitive period and explains the general exemptions from the CBAM.
- **Guidance 2** provides a quick guidance for the intended reader of this document, the operator of an installation producing CBAM goods. It provides a roadmap of the most important concepts of CBAM emissions monitoring and where to find more information in this series of documents.
- **Guidance 3** sets out the monitoring and reporting obligations and recommendations which are potentially applicable to any affected producer of CBAM goods.
- **Guidance 4** provides an overview of how the adjustment for free allocation is to be determined and what information is needed.
- **Guidance documents 5a, 5b, 5c, 5d, 5e and 5f** present an overview per sector of the production processes and value chains for goods that are included in the scope of the CBAM, as well as sector-specific monitoring and reporting considerations for each CBAM good where this is relevant. These sections are supplemented by examples for each sector.

The series of guidance documents is accompanied by an electronic template for information that should be used by installation operators to communicate information to the authorised CBAM declarants and CBAM Registry.

Presentation of numbers in EU documents

To align with EU legal documents, this guidance document uses the following convention when presenting numbers.

The decimal separator used to separate the integral part of a number from its fractional part is a comma, e.g.: 0,890

Thousands, and powers of 10^{3n} thereafter, are separated by a space, e.g.:

- fifteen thousand is written as 15 000
- fifteen million is written as 15 000 000

1.2 Reference regulations and legal acts

Several regulations and legal acts are referred to throughout this series of guidance documents. For ease of reading and to avoid ambiguity, simplified references are used when citing these acts.

- **CBAM Regulation:** Regulation (EU) 2023/956 and Regulation (EU) 2025/2083 amending Regulation (EU) 2023/956. Notably, where article numbers are given in this document without further specification, they always refer to the CBAM Regulation.

It can be accessed at the following link (consolidated version):

<http://data.europa.eu/eli/reg/2023/956/2025-10-20>

- **Methodology Act:** Commission Implementing Regulation (EU) 2025/2547, in legal texts referred to as “Implementing act pursuant to Article 7(7) of the CBAM Regulation”.

It can be accessed at the following link:

http://data.europa.eu/eli/reg_impl/2025/2547/oj

- **Default Values Act:** Commission Implementing Regulation (EU) 2025/2621, in legal texts referred to as “Implementing act pursuant to Article 7(7) of the CBAM Regulation”, also referred to as “default values determined in accordance with Annex IV to the CBAM Regulation”.

It can be accessed at the following

link: http://data.europa.eu/eli/reg_impl/2025/2621/oj

- **Verification Principles Act:** Commission Implementing Regulation (EU) 2025/2546, in legal texts referred to as “Implementing act pursuant to Article 8(3) of the CBAM Regulation”.

It can be accessed at the following link:

http://data.europa.eu/eli/reg_impl/2025/2546/oj

- **Accreditation & Verification Requirements Act:** Commission Delegated Regulation (EU) 2025/2551, in legal texts referred to as “Delegated act pursuant to Article 18(3) of the CBAM Regulation”.

It can be accessed at the following

link: http://data.europa.eu/eli/reg_del/2025/2551/oj

- **Free Allocation Adjustment Act:** Commission Implementing Regulation (EU) 2025/2620, in legal texts referred to as “Implementing act pursuant to Article 31(2) of the CBAM Regulation”.

It can be accessed at the following link:

http://data.europa.eu/eli/reg_impl/2025/2620/oj

For acronyms and definitions used in this document, please see **Annex B** and **Annex A**.

A series of icons are used throughout to help guide the reader:

Icon	Description of use
	Points to information of particular importance for operators of installations producing CBAM goods.
	Highlights simplified approaches of the general requirements of the CBAM.
	Used where recommended improvements are presented.
	Used where other documents, templates or electronic tools are available from other sources.
	Points to examples given for the topics discussed in the surrounding text.

2 CBAM GOODS AND PRODUCTION ROUTES

This section provides guidance on industry-sector-specific rules that apply for the definitive period (from 2026 onwards) in the fertilisers sector. It deals with the specification of products covered by the CBAM and the relevant production routes.

2.1 Foreword to sector-specific sections

The following sections provide an overview of the different production routes for the goods under fertilisers sector in CBAM and provide sector specific guidance. Where goods to which the same functional unit applies are produced in one installation via different production routes, the embedded emissions shall be determined in accordance with the averaging rules applicable in the definitive period.

Additional information on the production processes of the goods can also be found in the BREF¹ documents for best available techniques (BAT).

Diagrams used in the following sections.

For the system boundary graphics presented in the sections below, the following conventions are applied:

- **Production processes** (for which monitoring of the direct emissions would take place) are shown as **rectangles with white background**.
- **Optional** processes (e.g. CCS/CCU) are shown in **blue** boxes. In particular, CCS/CCU would not be taken into account for developing default values, but where you (as an operator) use them, the related emissions or emission savings should be taken into account for determining actual embedded emissions in line with the CBAM Regulation and the Methodology Act.
- **Materials** which are considered to have no embedded emissions are shown in **pink** boxes, materials with embedded emissions (precursor materials and final products, i.e. goods under the CBAM) in **green** boxes. Simple goods are shown in **normal** font, complex goods in **bold** font. Materials which are **optional** are shown in boxes with a **blue dashed** outline and must be included where actually used at the installation.
- Input materials are presented without trying to be complete. This means that the focus is on materials which are relevant for demonstrating the differences between different production routes. As a consequence, less important input materials and in particular fuels are usually omitted in order to keep graphs simple.
- **Electricity** as input is shown only in cases where it is the main “precursor” of the process (i.e. in particular for electric arc furnaces and electrolysis processes). Electricity is shown in **light grey** boxes.

¹ BAT Reference document (BREF), BAT being “Best Available Techniques” as defined by the IED (Industrial Emissions Directive). Relevant BREF documents are those for: the production of cement; for iron and steel production; large volume inorganic chemicals (which includes fertilisers); for Chlor-alkali; and for non-ferrous metals (which includes both aluminium and ferro-alloys). All BREFs can be found at the European IPPC Bureau, under <https://eippcb.jrc.ec.europa.eu/reference>.

2.2 Fertilisers sector goods and production routes

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I**, Table 1 “Mapping of CN codes to aggregated goods categories” (aggregated goods categories “Ammonia”, “Nitric acid”, “Urea” and “Mixed fertilisers”) to the Methodology Act.
 - **Annex I**, points 3.7–3.10 (“Ammonia”, “Nitric acid”, “Urea”, “Mixed fertilisers”), which define the functional unit, system boundaries and possible precursors for nitrogenous fertiliser goods, together with the cross-sectoral rules on emissions monitoring and attribution in **Annex II and Annex III**.
-

2.2.1 Unit of production and embedded emissions

The quantity of declared nitrogen containing fertiliser sector goods imported into the EU must be expressed using the functional unit applicable to the relevant fertiliser CBAM good (i.e. kilograms of nitrogen content or the relevant supplementary units, as applicable). As an operator, you must record the quantity of CBAM goods produced by installation or production process, for the purposes of reporting and calculating specific embedded emissions in accordance with the CBAM Regulation and the Methodology Act. For the purposes of calculating specific embedded emissions, the functional unit for fertiliser CBAM goods is defined in Article 4(4) of the Methodology Act, as kilograms of nitrogen contained in the goods produced under CN codes 2808 00 00, 2814 and 3105, and the relevant supplementary units laid down in Regulation (EEC) No 2658/87 for other fertiliser CN codes, which are defined as kilograms of nitrogen contained in the goods as well.

Industrial sector	Fertilisers
Production unit of goods	Functional unit as defined in Article 4(4) of the Methodology Act (kilograms of nitrogen contained in the goods for CN codes 2808 00 00, 2814 and 3105, or the relevant supplementary units laid down in Regulation (EEC) No 2658/87 for other fertiliser CN codes), reported separately for each type of CBAM good, by installation or production process in the country of origin.
Associated activities	Producing chemical precursors for nitrogenous fertiliser production, producing nitrogenous fertilisers by physical mixing or chemical reaction, and processing into their final form.
Relevant greenhouse gas emissions	Carbon dioxide (CO ₂) and nitrous oxide (N ₂ O)
Direct Emissions	Tonnes (metric) of CO ₂ e

Industrial sector	Fertilisers
Indirect emissions	Quantity of electricity consumed (MWh), source and emissions factor used to calculate the indirect emissions in tonnes (metric) of CO₂ or CO₂e. Indirect emissions resulting from electricity consumption must be monitored and taken into account in the embedded emissions of fertiliser CBAM goods, in accordance with Article 3(2) of the CBAM Regulation and the Methodology Act, since fertiliser CBAM goods are not listed in Annex II (“Goods for which only direct emissions are taken into account”).
Unit for embedded emissions	Tonnes CO ₂ e emissions per functional unit of goods (kilogram of nitrogen contained in the fertiliser), reported separately for each type of goods, by installation in the country of origin.

During the transitional period, the fertiliser industry sector reported both direct and indirect emissions, with indirect emissions reported separately. In the definitive period, fertiliser CBAM goods are subject to the general rule in Article 3(2) of the CBAM Regulation: both direct and indirect emissions, as well as the embedded emissions of any precursors, must be taken into account when determining embedded emissions, since fertiliser CBAM goods are not listed in Annex II (“Goods for which only direct emissions are taken into account”). Emissions must be reported in metric tonnes CO₂ equivalent (t CO₂e) per functional unit of fertiliser good (per kg of nitrogen content), calculated for the specific installation or production process in your country of origin. Where fertiliser goods are produced and placed on the market with different nitrogen contents, emissions shall be calculated and averaged in line with the composition-based rules set out in Annex III, Section B to the Methodology Act (i.e. by averaging across nitrogen-content ranges, with ranges not exceeding 10%, see equation 66).

The following sections set out how the system boundaries of fertiliser sector goods should be defined and identify elements of the production process that must be included for the purposes of monitoring and reporting.

2.2.2 Definition and explanation of sector CBAM goods covered

The following Table 2-1 lists the relevant goods in scope for the CBAM definitive period in the fertiliser industry sector. The CN codes in the central column are the starting point for which ‘production processes’ are to be defined for the purpose of monitoring at installation level and, where multiple CN codes share the same precursors and production steps, may be grouped within one production process, in line with Annex I, Table 1 to the Methodology Act.

Table 2-1: CBAM goods in the fertiliser sector²

Aggregated goods category	Product CN Code ³	Description
Nitric acid	2808 00 00	Nitric acid; sulphonitric acids
Urea	3102 10 ⁴	Urea, whether or not in aqueous solution
Ammonia	2814	Ammonia, anhydrous or in aqueous solution
Mixed fertilisers	2834 21 00, 3102, 3105 - Except 3102 10 (Urea) and 3105 60 00	2834 21 00 – Nitrates of potassium 3102 – Mineral or chemical fertilisers, nitrogenous - Except 3102 10 (Urea) 3105 – Mineral or chemical fertilisers containing two or three of the fertilising elements nitrogen, phosphorus, and potassium; other fertilisers - Except: 3105 60 00 – Mineral or chemical fertilisers containing the two fertilising elements phosphorus and potassium ⁵

The aggregated goods categories listed in Table 2-1 include both the finished nitrogenous fertiliser goods and chemical precursor goods (intermediate products) that are consumed in the production of fertiliser.

The system boundaries specified in the Methodology Act do not mention explicitly which precursors are relevant for the production process. Therefore, Table 2-2 should help to identify the possible precursors by aggregated goods category and production route.

² Source: Annex I to the CBAM Regulation; Annex I (“Mapping of CN codes to aggregated goods categories”) to the Methodology Act.

³ Where the CN codes indicated in this table consist of 4, 5 or 6 digits, all 8-digit CN codes sharing the same initial digits shall be considered included, unless otherwise specified.

⁴ The urea six-digit code includes at eight-digit CN code level: 3102 10 12; 3102 10 15; 3102 10 19; 3102 10 90

⁵ Only nitrogen (N) containing fertilisers have significant embedded emissions, therefore their precursors are included in the CBAM.

Table 2-2: Aggregated goods categories, their production routes and possible precursors

Aggregated Goods Category	Precursors
<i>Production route</i>	
Ammonia	Hydrogen ⁶ .
<i>Haber Bosch with steam reforming</i>	
<i>Haber Bosch with gasification</i>	
Nitric Acid	Ammonia (as 100% ammonia).
Urea	Ammonia (as 100% ammonia).
Mixed fertiliser	If used in the process: ammonia (as 100% ammonia), nitric acid (as 100% nitric acid), urea, mixed fertilisers (in particular salts containing ammonium or nitrate).

For the production of mixed fertiliser, not all precursors will apply in every case. Note in particular that in some instances an aggregated goods category (mixed fertiliser itself) may be used as a precursor for its own category, depending on the final formulation of the mixed fertiliser product required.

The final nitrogenous chemical fertiliser goods produced from the precursors (in bulk in integrated plants) are defined as complex goods as they include the embedded emissions from precursor goods.

The production of fertiliser sector goods is by a number of different process routes, outlined below.

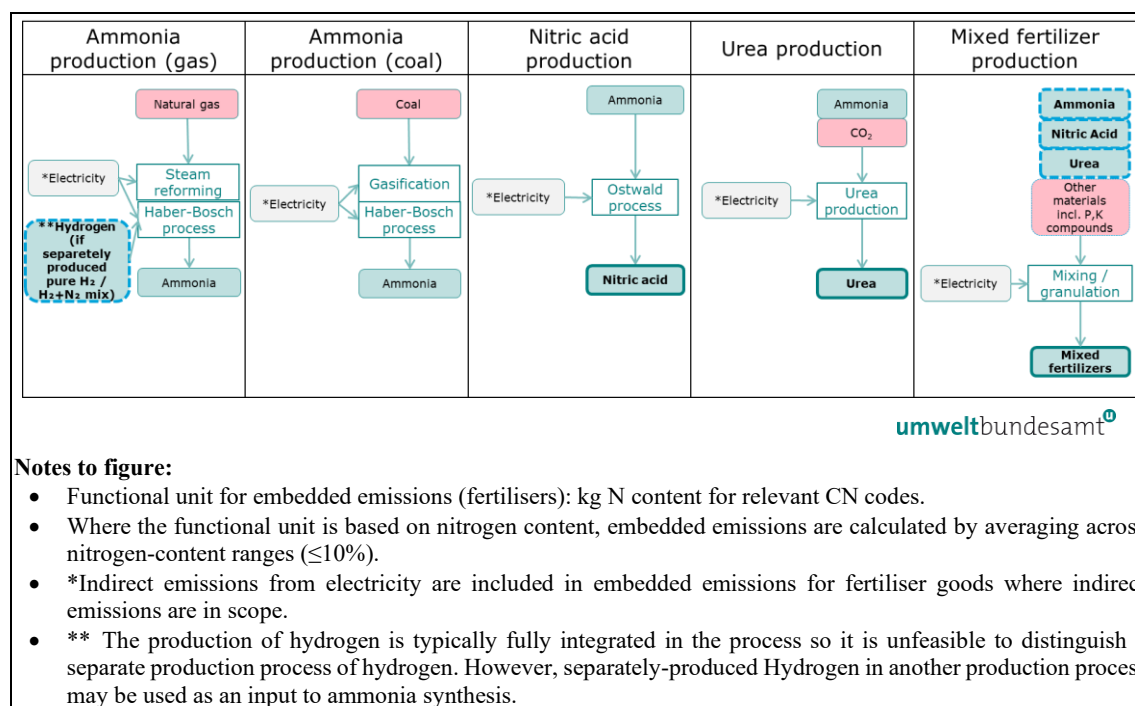
2.2.3 Definition and explanation of relevant production processes and routes

The system boundaries of chemical precursors and fertilisers are distinct and may, under certain conditions, be added together to include all processes directly or indirectly linked to the production processes for these goods, including input activities to the process, and output activities from the process in line with the definitions of functional units, system boundaries and precursors for “Ammonia”, “Nitric acid”, “Urea” and “Mixed fertilisers” in Annex I, points 3.7–3.10, and the rules for attributing emissions to goods in Annex III of the Methodology Act.

⁶ Where hydrogen is used in the process, it shall be treated as a precursor with its own embedded emissions determined in accordance with the rules applicable to the hydrogen aggregated goods category. Indirect emissions of electricity associated with the hydrogen precursor may be disregarded when determining the embedded emissions of ammonia CBAM goods. However, this applies only where hydrogen from external installations or separate production processes (e.g. an electrolysis process on the ammonia production site) is used in the synthesis of ammonia. In cases of generic Haber-Bosch processes, where the production of hydrogen is only a highly integrated intermediate step of the overall production process (and indirect emissions from hydrogen production can thus not be determined separately), the indirect emissions of the complete process are to be included.

The following Figure 2-1 provides an overview of the different processes and process routes for the production of nitrogenous fertiliser and its precursors.

Figure 2-1: System boundaries and value chain for the production of nitrogenous fertiliser and its precursors – overview



Urea is used as a precursor in mixed fertiliser production but may also be used as a convenient fertiliser on its own due to its high nitrogen content and is therefore both a CBAM good in its own right (aggregated goods category “Urea”) and, where applicable, a precursor for mixed fertilisers as defined in Annex I, Table 1 of the Methodology Act.

Mixed fertilisers comprise all kinds of nitrogen (N) containing fertilisers, including ammonium nitrate, calcium ammonium nitrate, ammonium sulphate, ammonium phosphates, urea ammonium nitrate solutions, as well as nitrogen-phosphorus (NP), nitrogen-potassium (NK) and nitrogen-phosphorus-potassium (NPK) fertilisers falling under the “Mixed fertilisers” aggregated goods category of the Methodology Act. In addition, the amounts of different nitrogen compounds contained in the final product shall be recorded in accordance with Annex I, point 3.10.1 of the Methodology Act.

2.2.3.1 Ammonia – Haber-Bosch with steam reforming production route

Ammonia is synthesised from nitrogen and hydrogen via the Haber-Bosch process. Hydrogen for the process is obtained in this production route by steam reforming natural gas or biogas, whilst nitrogen is obtained from the air. The overall reaction is highly endothermic, and process heat is supplied by the combustion of natural gas or other gaseous fuel. Any carbon monoxide produced is almost all converted to carbon dioxide. The production of hydrogen is typically fully integrated in the process, so it is unfeasible to distinguish a separate production process of hydrogen. On some occasions, hydrogen produced from other separated production processes can be also used and its embedded emissions shall be accounted.

The Methodology Act and its annexes (in particular the provisions for the “Ammonia” aggregated goods category) define the system boundaries for direct emissions monitoring for the Haber–Bosch process with steam reforming production route. For the purposes of this guidance, those system boundaries can be described as encompassing:

“ – All processes directly or indirectly linked to the production process, and from flue gas cleaning.

– All fuels shall be monitored, irrespective of whether used as energetic or non-energetic input.

– Where biogas is used, the provisions of Section B.3.3 of Annex II shall be applied.”

If hydrogen used in the process is separately produced, it is a precursor with its own embedded direct emissions.

Indirect emissions that result from electricity consumed by the ammonia production process must be monitored and taken into account in embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

In line with the above definition of system boundaries, the following production steps may be regarded as being within the system boundaries of the Haber-Bosch with steam reforming process:

- Steam reforming natural gas or biogas.
- Synthesis of ammonia – from hydrogen and nitrogen, at high temperature and pressure in the presence of a catalyst; ammonia condensation, purification and storage (if applicable).
- Emissions control – for treating releases to air, water or ground.

The stream of carbon dioxide from the production of ammonia is of high purity and can be separated, captured and transferred elsewhere for other uses e.g. for urea production.

Note that the ammonia produced is reported as 100% ammonia, whether in hydrous or anhydrous form.

2.2.3.2 Ammonia – Haber-Bosch with gasification production route

With this production route, hydrogen is obtained by the gasification of hydrocarbons, typically from heavy feedstocks such as coal, heavy refinery fuels or other fossil feedstocks. A synthesis gas containing hydrogen is produced, which has to be purified before it can be used for the next production step. Ammonia is then synthesised from the hydrogen produced and from nitrogen obtained from the air, at high temperature and pressure in the presence of a catalyst. Any carbon monoxide produced is almost all converted to carbon dioxide. The production of hydrogen is typically fully integrated in the process so it is unfeasible to distinguish a separate production process of hydrogen. In some occasion, hydrogen produced from other separated production processes can be also used and its embedded emissions shall be accounted.

The Methodology Act and its annexes (in particular the provisions for the “Ammonia” aggregated goods category) also define the system boundaries for direct emissions

monitoring for the Haber–Bosch process with gasification production route. For the purposes of this guidance, those system boundaries can be described as encompassing:

“ – All processes directly or indirectly linked to the production processes, and from flue gas cleaning.

– Each fuel input shall be monitored as one fuel stream, irrespective of whether it is used as energetic or non-energetic input.”

If hydrogen used in the process is separately produced, it is a precursor with its own embedded direct emissions.

Indirect emissions that result from electricity consumed by the ammonia production process must be monitored and taken into account in embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

In line with the above definition of system boundaries, the following production steps may be regarded as being within the system boundaries of the Haber-Bosch with gasification process:

- Gasification (partial oxidation).
- Synthesis of ammonia – from hydrogen and nitrogen, at high temperature and pressure in the presence of a catalyst; ammonia condensation, purification and storage (if applicable).
- Emissions control – for treating releases to air, water or ground.

Note that the ammonia produced is reported as 100% ammonia, whether in hydrous or anhydrous form.

2.2.3.3 Nitric acid (and sulphonitric acids) production process

Nitric acid is mostly produced via the oxidation of ammonia by Ostwald process. Ammonia is first oxidised in the presence of a catalyst to form nitrogen oxide, which is then further oxidised to nitrogen dioxide, followed by absorption in water in an absorption tower to form nitric acid. The reaction is exothermic and heat and power may be recovered to the process.

The Methodology Act and its annexes (in particular the provisions for the “Nitric acid” aggregated goods category) define the system boundaries for direct emissions monitoring for the nitric acid production route. For the purposes of this guidance, those system boundaries can be described as encompassing:

“ – All processes directly or indirectly linked to the production processes, and from flue gas cleaning.

– All sources emitting N₂O from the production process, including unabated and abated emissions. Any N₂O emissions from the combustion of fuels are excluded from monitoring.”

A precursor is ammonia (as 100% ammonia). Indirect emissions that result from electricity consumed by the production process must be monitored and taken into account in embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of the nitric acid process production:

- Raw material preparation – evaporation and filtration of ammonia and process air.
- Oxidation of ammonia – to nitrogen oxide, all process steps.
- Further oxidation and absorption – to nitrogen dioxide and absorption in water to form nitric acid, all process steps.
- Emissions control – for treating releases to air, water or ground.

Note that nitric acid produced is reported as 100% nitric acid.

2.2.3.4 Urea production process

Urea is synthesised by reacting ammonia and carbon dioxide together at high pressure, to form ammonium carbamate, which is then dehydrated to form urea.

The Methodology Act and its annexes (in particular the provisions for the “Urea” aggregated goods category) define the system boundaries for direct emissions monitoring for the urea production route. For the purposes of this guidance, those system boundaries can be described as encompassing:

“ – All processes directly or indirectly linked to the production processes, and from flue gas cleaning.

– Where CO₂ is received from another installation as process input, the CO₂ received shall be considered an emission, if not already counted as emission of the installation where the CO₂ was produced.”

A precursor is ammonia (as 100% ammonia). Indirect emissions that result from electricity consumed by the production process must be monitored and taken into account in embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of the urea production process:

- Raw material preparation – evaporation and filtration of ammonia, CO₂.
- Production of urea – all process steps, from synthesis to particle formation.
- Emissions control – for treating releases to air, water or ground.

The ammonia and CO₂ consumed by this production process are usually delivered from other production processes on the same site.

2.2.3.5 Mixed fertilisers production process

A wide range of operations are included in the production of all kinds of nitrogen containing mixed fertilisers (especially ammonium salts and NP, NK and NPK), such as

mixing, neutralisation⁷, particle formation (such as by granulation or prilling), irrespective of whether only physical mixing or chemical reactions take place.

The Methodology Act and its annexes (in particular the provisions for the “Mixed fertilisers” aggregated goods category) define the system boundaries for direct emissions monitoring for the mixed fertiliser production route. For the purposes of this guidance, those system boundaries can be described as encompassing:

“ – All processes directly or indirectly linked to the production processes, such as driers and for heating input materials, and from flue gas cleaning.

Precursors (if used in the process) are: ammonia (as 100% ammonia); nitric acid (as 100% nitric acid); urea; mixed fertilisers (in particular salts containing ammonium or nitrate). Indirect emissions that result from electricity consumed by the production process must be monitored and taken into account in embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of the mixed fertiliser production process:

- Raw material preparation.
- Production of mixed fertiliser – all process steps.
- Emissions control – for treating releases to air, water or ground.

3 FERTILISERS SECTOR-SPECIFIC REQUIREMENTS

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I, point 3** – Functional unit and system boundaries. Sub-sections 3.7 to 3.10 (fertiliser sector aggregated goods categories).
 - **Annex II, Section B.6** Requirements for a measurement-based methodology for CO₂ and N₂O. Section **B.8**. Requirements for CO₂ transfers between installations. Section **B.9.3** Additional rules for determining emissions from nitric acid production, covering: **B.9.3.1** General rules for measurement-of N₂O emissions; **B.9.3.2** Determination of flue gas flow; **B.9.3.3** Oxygen (O₂) concentrations.
 - **Annex III: Section A.2.1**, for attribution rules for co-products in the case of chemicals and fertilisers.
 - **Annex IV, point 2** – Sector-specific parameters to be included in the operator’s emissions report.
-

⁷ Nitrogen containing chemical fertilisers are produced by the neutralisation of an acid with ammonia to form the corresponding ammonium salt. Fertilisers produced in this way include ammonium nitrate, calcium ammonium nitrate, ammonium sulphate, ammonium phosphates, urea ammonium nitrate.

3.1 Sector-specific requirements for monitoring and reporting

Direct and indirect embedded emissions must be monitored in line with the methodology set out in the Methodology Act and outlined in this of the guidance document.

3.1.1 Emissions monitoring

Relevant emissions that must be monitored and reported for the fertilisers sector are:

- Carbon dioxide emissions (direct) from the fuel combustion process, from stationary plant only (excludes emissions from any mobile plant such as vehicles).
- Carbon dioxide and nitrous oxide (N₂O) emissions (direct) from the process, in particular:
 - In the production of nitric acid, N₂O emissions from the catalytic oxidation of ammonia and/or from the NO_x/N₂O abatement units (but not from combustion); and
 - Under certain conditions CO₂ transferred from the production process for ammonia to other installations. However, CO₂ consumed in the production of urea has to be considered as being emitted at the installation, as it does not comply with the requirements for permanently chemically bound CO₂.
- Carbon dioxide emissions (direct) resulting from the production of measurable heating (e.g. steam) and cooling that is consumed within the system boundaries of the production process, regardless of the location of the production of the heat (i.e. from on-site generation or from imports from off-site).
- Carbon dioxide emissions (direct) resulting from emissions control (e.g. from carbonate raw materials such as soda ash used for acidic flue gas cleaning). This is included for any good where this is applicable.

Direct emissions from the different source streams above are not reported separately but are added together to result in the total direct emissions for the installation or production process.

Indirect emissions from electricity consumed must be reported separately from direct emissions. In accordance with section 3.1 of Annex I to the Methodology Act, indirect emissions of precursors do not have to be included if those precursors are listed in Annex II to the CBAM Regulation, i.e. if indirect emissions are excluded for those precursors. This is relevant for ammonia production, where hydrogen can be a precursor. Hydrogen's indirect emissions are not counted even though ammonia as a good has to take into account indirect emissions. However, this needs to be understood to apply **only where hydrogen from external installations or separate production processes is used** in the synthesis of ammonia. In cases of generic Haber-Bosch processes, where the production of hydrogen is only a highly integrated intermediate step of the overall production process, the indirect emissions of the complete process are to be included.

Note that other N₂O emissions resulting from the combustion of fuels are excluded from the system boundaries.

3.1.2 Additional rules

Functional unit and Attribution of emissions for mixed fertilisers

For fertiliser goods whose **functional units** are **kilograms of nitrogen content** and that are sold in different ranges of composition, you as the operator will need to calculate the specific embedded emissions of the goods according to the nitrogen contained in the goods averaged for each range of composition, using the following equation.

$$SEE_{g(Ni)} = SEE_g \cdot N_i \quad (\text{Equation 65})$$

Where:

$SEE_{g(Ni)}$ is the specific embedded emissions of the good with nitrogen content N_i expressed as t CO₂e per tonne of good;

SEE_g is the specific embedded emissions calculated in Equation 59 or 62, i.e. expressed as t CO₂e per kg N, and

N_i is the average nitrogen content of the goods within a range of composition in kilograms of nitrogen per tonnes of goods.

Note that for goods whose functional units are kg of nitrogen content, and that are sold in custom-made compositions at the request of the client, you as the operator are required to issue a declaration of nitrogen content for each consignment and with the calculation of specific embedded emissions according to the correspondent Equation 65, where N_i will be the specific nitrogen content of the consignment⁸.

Measurable heat produced from exothermic chemical processes

Where an installation consumes measurable heat produced/recovered from an exothermic chemical process other than combustion, such as in the production of ammonia or nitric acid, the amount of recovered heat consumed is determined separately from other measurable heat, and is assigned zero CO₂ emissions.

Production of electricity

If electricity is produced within the production process, a correction to the attributed emissions has to be made. Where the electricity stems from processes without combustion (e.g. expansion turbines in ammonia production), the emission factor of that electricity is considered zero.

Measurement-based approach for monitoring N₂O emissions

If you, as an operator, produce nitric acid, you must monitor N₂O emissions from the process (not from combustion) using a Continuous Emission Measurement System (CEMS) installed at a suitable measurement point⁹.

⁸ In mixed fertiliser production European fertiliser legislation requires the content of N (in its different forms, as ammonium (NH₄⁺) or nitrate (NO₃⁻), urea or other (organic) forms) to be clearly indicated on the packaging or in accompanying sales documentation in case of bulk deliveries. These content values may be used to determine embedded emissions of any mixed fertiliser.

⁹ If there are multiple emissions points that cannot be monitored from a single location, emissions from these various points must be separately monitored and the results combined for reporting purposes.

N₂O emissions are only regarded relevant for monitoring in the production of nitric acid. However, where nitric acid or the resulting nitrates (mixed fertilisers) are used as precursor, the related N₂O emissions form an integral part of the embedded emissions, which are expressed as t CO₂e:

$$CO_{2(e)} [t] = N_2O_{annual}[t] \times GWP_{N_2O} \text{ (Equation 18)}$$

Where:

N_2O_{annual} ... total annual N₂O emissions.

GWP_{N_2O} ... Global Warming Potential of N₂O (t CO₂e / t N₂O). Please see Annex II point G to the Methodology Act for relevant GWP values as shown in Table 3-1.

Table 3-1: Global warming potentials

Gas	Global warming potential
N ₂ O	265 t CO ₂ e / t N ₂ O
CF ₄	6 630 t CO ₂ e / t CF ₄
C ₂ F ₆	11 100 t CO ₂ e / t C ₂ F ₆

For determining the flow rate of the flue gas, the Methodology Act states that the mass balance method is preferred over continuous emissions flow measurements.

3.1.3 Additional reporting requirements

Table 3-2 lists out the additional information that needs to be provided by you as an operator to importers or operators using your goods as precursors, in your verified emissions report which you provide to them.

Table 3-2: Additional fertiliser sector parameters requested in the CBAM declaration

Aggregated category	good	Reporting requirement in the annual report
Ammonia ¹⁰		– Concentration, if hydrous solution. – Content of nitrogen (N).
Nitric acid ¹¹		– Concentration (mass %).

¹⁰ Both hydrous and anhydrous ammonia shall be reported jointly as 100% ammonia.

¹¹ Amounts of nitric acid produced shall be monitored and reported as 100% nitric acid.

Aggregated category	good	Reporting requirement in the annual report
		– Content of N.
Urea		- Purity (mass % urea contained, % N contained). - Content of N.
Mixed fertilisers ^{12,13}		- Information required anyway under Regulation (EU) 2019/1009: - Content of N as ammonium (NH₄⁺); - Content of N as nitrate (NO₃⁻); - Content of N as urea; - Content of N in other (organic) forms. - Content of N total.

You need to ensure that you collect all the parameters necessary for your CBAM goods and communicate them to the importers of your goods. The importer will need to report the additional parameters in their annual CBAM declaration for the relevant reporting period.

4 WORKED EXAMPLE FOR THE FERTILISER SECTOR

The following worked example shows how specific embedded emissions are derived for a particular mixed fertiliser grade, NPK 15-15-15, produced by mixing and granulation.

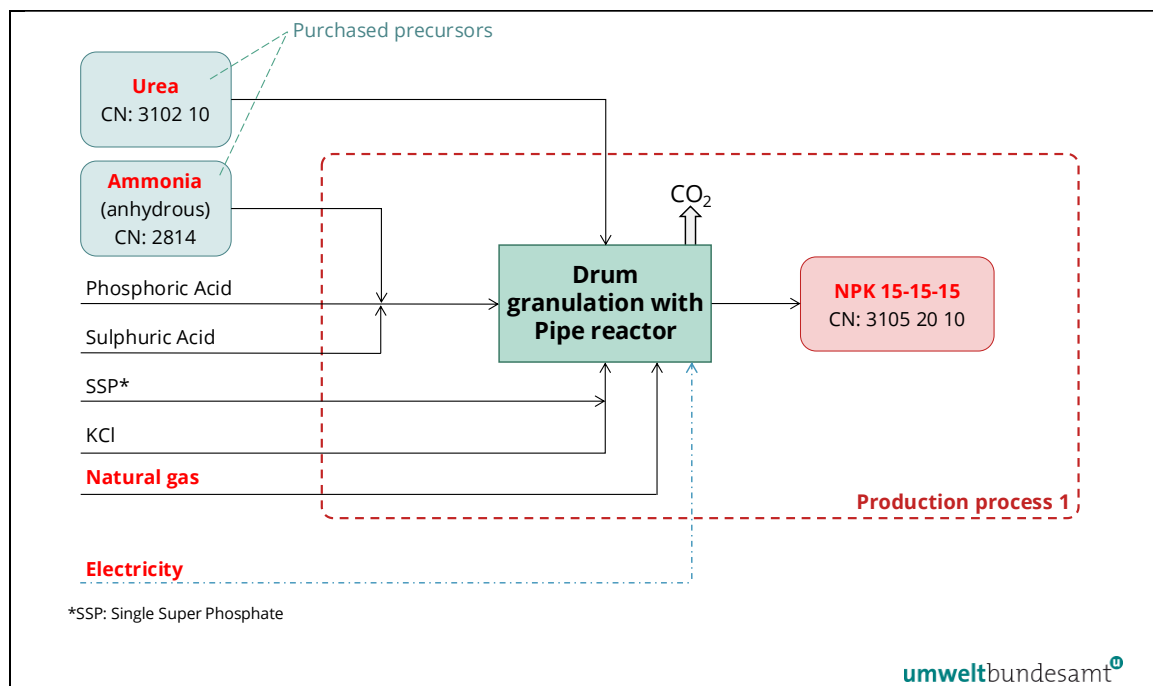
The resulting embedded emissions of imports into the EU are then calculated at the end of the example for reporting in the definitive period.

Figure 4-1 gives an outline view of the installation and shows the system boundaries as a hatched line for the single production process. The physical units carrying out the production process have been grouped under ‘Granulation with Pipe reactor’ (including an assumed dryer using natural gas) and the inputs and outputs and sources of emissions have been identified.

¹² The amounts of different nitrogen compounds contained in the final product must be recorded in accordance with Regulation (EU) 2019/1009 laying down rules on the making available on the market of EU fertilizing products.

¹³ Regulation (EU) 2019/1009 of the European Parliament and of the Council laying down rules on the making available on the market of EU fertilizing products.
See: <http://data.europa.eu/eli/reg/2019/1009/2025-12-30>

Figure 4-1: Fertiliser example – Overview and complete monitoring approach for the production of a mixed fertiliser grade.



Inputs into the production process are raw materials, precursor goods urea and ammonia (anhydrous), and electrical energy. Outputs are mixed fertiliser product.

The inputs and outputs highlighted in red text above are the parameters that would need to be monitored by the operator in order to attribute emissions and determine direct and indirect specific embedded emissions for the production process.

The direct and indirect emissions that are monitored in this example result from:

- Direct emissions from natural gas used in the dryer.
- Indirect emissions from the electrical energy consumed by the production process.

The inputs of precursors (with embedded emissions) and the activity level of the mixed fertiliser goods produced also need to be monitored.

Note that a single mixed fertiliser production process can make a wide range of different fertiliser grades (or formulations) using different quantities of precursor. However, this issue was resolved in the rules for the definitive phase by introduction of the ‘functional unit’. In this way, the specific embedded emissions for each fertiliser grade can be determined easily.

The following needs to be monitored:

- The relevant mass of each precursor used over the reporting period;
- The production ‘recipe’ of each mixed fertiliser grade (or an analysis of the final product); and
- The specific embedded emissions of the precursors used to make a given mixed fertiliser grade.

- Assuming that the granulation and drying process is similar for all produced fertiliser grades, the direct and indirect emissions of the production process can be monitored over the full reporting period and then divided by the total activity level of the process, i.e. the total amount of all fertilisers produced in the reporting period. This gives a value of energy per tonne of fertiliser as used in the calculation in Table 4-1.
- The embedded emissions are first calculated per functional unit (kg N used in the reporting period) and then converted into embedded emissions per tonne of (mixed) fertiliser using equation 65 (see section 3.1.2).

For the example, it is assumed that the installation is situated in Türkiye. The electricity grid emission factor as found in Annex II to the default Values Act is used for the calculation of indirect embedded emissions. The precursors ammonia and urea are assumed to be sourced from Azerbaijan. Production year both for the precursors and the actual production are assumed to be 2026. Default values including mark-up are taken from Annex I to the Default Values Act. Table 4-1 sets out the process by which total direct and indirect specific embedded emissions for mixed fertiliser product NPK 15-15-15, are determined.

Total direct and indirect specific embedded emissions for the mixed fertiliser product are calculated by combining the default SEE values for precursors (which cover both direct and indirect emissions) and for energy needed for granulation, per tonne product, as above

Precursor goods are NH₃ and Urea. In order to determine the total embedded emissions of the mixed fertiliser product, the amount (kg) of each precursor used per tonne of mixed fertiliser product is taken into account, e.g. for urea the total input mass of precursor per tonne of product is 160 kg:

- Ammonia total embedded emissions: $0,093 \text{ t} / \text{t} \times 2,273 \text{ t CO}_2 / \text{t} = \mathbf{0,211 \text{ t CO}_2 / \text{t}}$ mixed fertiliser product.
- Urea total embedded emissions: $0,160 \text{ t} / \text{t} \times 1,545 \text{ t CO}_2 / \text{t} = \mathbf{0,247 \text{ t CO}_2 / \text{t}}$ mixed fertiliser product.

Table 4-1: Example calculation of total direct and indirect specific embedded emissions for NPK mixed fertiliser.

Inputs	Input mass (kg / t)	Precursor embedded emissions (t CO ₂ /t)		Embedded emissions (t CO ₂ /t)	
		Direct / total	indirect	Direct / total	indirect
KCl	251,3	n.a.	n.a.	n.a.	n.a.
SSP ¹⁴ 17% P ₂ O ₅	200,0	n.a.	n.a.	n.a.	n.a.
Phosphoric acid (40% P ₂ O ₅)	300,0	n.a.	n.a.	n.a.	n.a.
Sulphuric acid (96 wt-%)	116,0	n.a.	n.a.	n.a.	n.a.
NH ₃ ¹⁵	93,0	2,273	(included) ¹⁶	0,211	--
Urea ¹⁷	160,0	1,545	(included) ¹⁶	0,247	--
Emissions from energy needed for granulation (average of reporting period)				0,018	0,0126
Total SEE for mixed fertiliser product NPK 15-15-15				0,489	
Total SEE expressed per functional unit, i.e. t CO₂e/kg N				0,00326	

Direct and indirect emissions arising from the mixing and granulation production process must also be included, as is done in Table 4-1 above per tonne of product.

Other chemical raw material inputs (KCl, SSP, Phosphoric and Sulphuric acids) do not have embedded emissions and do not need to be taken into account.

Note on calculation pathway: In the example above, only a single NPK formulation is produced. Therefore, it is simple to make all calculations per tonne of final fertiliser as presented above. However, if the installation produces many different NPK grades, the approach would be different:

- Instead of the specific input of each fertiliser component (precursor and other inputs) expressed as kg/t, the table would have to contain the total amount of each component (in tonnes) used during the production year.
- As the activity level, the total amount of nitrogen contained in all produced NPKs would be calculated (expressed as kg N, not differentiating between ammonia and urea inputs).
- Then an average level of embedded emissions per NPK would be calculated, expressed as t CO₂e/ kg N.

¹⁴ Single Super Phosphate

¹⁵ CN 2814 10 00 – Anhydrous ammonia

¹⁶ The default values include the indirect embedded emissions, where relevant. This is the case for ammonia and urea.

¹⁷ CN 3102 10 19 – Urea, whether or not in aqueous solution, containing >45% nitrogen in relation to the weight of the dry product

- Finally, for each NPK the embedded emissions per t would be calculated from the value per kg N and the N content.

Using the above approach, the CBAM reporting obligation due for the import of mixed fertiliser product into the EU during the definitive period may then be determined; for example, for the import of 100 tonnes of NPK 15-15-15 product:

- **Definitive period:**

- Total embedded emissions = $100 \text{ t} \times 0,489 \text{ t CO}_2 / \text{t} = 48,9 \text{ t CO}_2$

Annex A

List of definitions

Term	Definition
‘Accuracy’	means the closeness of the agreement between the result of a measurement and the true value of the particular quantity or a reference value determined empirically using internationally accepted and traceable calibration materials and standard methods, taking into account both random and systematic factors
‘Activity data’	means the amount of fuels or materials consumed or produced by a process relevant for the calculation-based methodology, expressed in terajoules (TJ), mass in tonnes or (for gases) volume in normal cubic metres, as appropriate
‘Actual emissions’	means the emissions calculated based on primary data from the production processes of goods and from the production of electricity consumed during those processes as determined in accordance with the methods set out in Annex IV [to the Methodology Act]
‘Activity level’	means the quantity of goods to which the same functional unit applies that are produced within the system boundaries of a production process during a reporting period
‘Aggregated goods categories’	means aggregated goods categories pursuant to Table 1 of point 2 of Annex I. A group of CBAM goods with their CN codes, used for the purpose of defining system boundaries of production processes.
‘Agricultural, aquaculture, fisheries and forestry residues’	means residues that are directly generated by agriculture, aquaculture, fisheries and forestry and that do not include residues from related industries or processing
‘Authorised CBAM declarant’	means a person authorised by the competent authority in accordance with Article 17 [of the CBAM Regulation]. The ‘authorised CBAM declarant’ [which replaces the term ‘reporting declarant’ from the transitional period] may be importers of CBAM goods, or their indirect customs representatives, who have applied for and been granted the status of authorised CBAM declarant.
‘Batch’	means an amount of fuel or material representatively sampled and characterised, and transferred as one shipment or continuously over a specific period of time
‘Biomass’	means biomass as defined in Article 2, point (24) of the Renewable Energy Directive (RED II) ¹⁸ ; it includes bioliquids and biofuels as defined in Article 2, point (32) and (33), biomass fuels as defined in Article 2, point (27) and biogas as defined in Article 2 point (28) of the RED II;

¹⁸ Directive (EU) 2018/2001

Term	Definition
‘Biomass fraction’	means the ratio of carbon stemming from biomass to the total carbon content of a fuel or material, expressed as a fraction
‘Calculation factors’	means net calorific value, emission factor, preliminary emission factor, oxidation factor, conversion factor, carbon content or biomass fraction
‘Calibration’	means the set of operations, which establishes, under specified conditions, the relations between values indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material and the corresponding values of a quantity realised by a reference standard
‘Carbon price’	means the monetary amount paid in a third country, under a carbon emissions reduction scheme, in the form of a tax, levy or fee or in the form of emission allowances under a greenhouse gas emissions trading system, calculated on greenhouse gases covered by such a measure, and released during the production of goods
‘CBAM certificate’	means a certificate in electronic format corresponding to one tonne of CO ₂ e of embedded emissions in goods
‘CO₂ emission factor’	means the weighted average of the CO ₂ intensity of electricity produced from fossil fuels within a geographic area. The CO ₂ emission factor is the result of the division of the CO ₂ emission data of the electricity sector by the gross electricity generation based on fossil fuels in the relevant geographic area. It is expressed in tonnes of CO ₂ per megawatt-hour
‘Combined nomenclature’ (CN)	means the classification of goods, designed to meet the needs of: i) the Common customs tariff, setting import duties for products imported into the European Union (EU), as well as the Integrated tariff of the European Communities (Taric), incorporating all EU and trade measures applied to goods imported into and exported out of the EU; ii) the international trade statistics of the EU. The CN provides the means of collecting, exchanging and publishing data on EU international trade statistics. It is also used for the collection and publication of international trade statistics in intra-EU trade.¹⁹
‘Combustion emissions’	means greenhouse gas emissions occurring during the exothermic reaction of a fuel with oxygen
‘Competent authority’	means the authority designated by each Member State in accordance with Article 11 [of the CBAM Regulation]

¹⁹ For definition see: [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_\(CN\)](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_(CN))

Term	Definition
‘Continuous emission measurement’ (CEM)	means a set of operations having the objective of determining the value of a quantity by means of periodic measurements, applying either measurements in the stack or extractive procedures with a measuring instrument located close to the stack, whilst excluding measurement methodologies based on the collection of individual samples from the stack
‘Control system’	means the operator’s risk assessment and entire set of control activities, including the continuous management thereof, that an operator has established, documented, implemented and maintained pursuant to point A.2. of Annex II
‘Complex goods’	means goods other than simple goods
‘Conservative’	means that a set of assumptions is defined in order to ensure that no under-estimation of reported emissions or over-estimation of production of heat, electricity or goods occurs
‘Conversion factor’	means the ratio of carbon emitted as CO ₂ to the total carbon contained in the source stream before the emitting process takes place, expressed as a fraction, considering CO emitted to the atmosphere as the molar equivalent amount of CO ₂
‘Co-product’	means any of two or more products resulting from the same production process
‘Customs declarant’	means a declarant as defined in Article 5, point (15) of Regulation (EU) No 952/2013 lodging a customs declaration for release for free circulation of goods in its own name or the person in whose name such a declaration is lodged
‘CCUS system’	means a group of economic operators with technically connected installations and transport equipment for CO ₂ capture, transport, use in production of goods, or geological storage
‘Data flow activities’	mean activities related to the acquisition, processing and handling of data that are needed to draft an emissions report from primary source data
‘Data set’	means one type of data, either at installation level or production process level as relevant in the circumstances, as any of the following: (a) the amount of fuels or materials consumed or produced by a production process as relevant for the calculation-based methodology, expressed in terajoules, mass in tonnes, or for gases as volume in normal cubic metres, as appropriate, including for waste gases; (b) a calculation factor;

Term	Definition
	(c) net quantity of measurable heat, and the relevant parameters required for determining this quantity, in particular: i) mass flow of heat transfer medium; and ii) enthalpy of transmitted and returned heat transfer medium, as specified by composition, temperature, pressure and saturation; (d) quantities of non-measurable heat, specified by the relevant quantities of fuels used for producing the heat, and the net calorific value (NCV) of the fuel mix; (e) quantities of electricity; (f) quantities of CO₂ transferred between installations; (g) quantities of precursors received from outside the production process, and their relevant parameters, such as country of origin, used production route, specific direct and indirect emissions;
‘Default value’	means a value, which is calculated or drawn from secondary data, which represents the embedded emissions in goods
‘Direct emissions’	means emissions from the production processes of goods including emissions from the production of heating and cooling that is consumed during the production processes, irrespective of the location of the production of the heating or cooling
‘Embedded emissions’	means direct emissions released during the production of goods and indirect emissions from the production of electricity that is consumed during the production processes, calculated in accordance with the methods set out in Annex IV and further specified in the implementing acts adopted pursuant to Article 7(7)
‘Emissions’	mean the release of greenhouse gases into the atmosphere from the production of goods
‘Emission factor’	means the average emission rate of a greenhouse gas relative to the activity data of a source stream assuming complete oxidation for combustion and complete conversion for all other chemical reactions
‘Emission factor’ for electricity	means the default value, expressed in CO ₂ e, representing the emission intensity of electricity consumed in production of goods.
‘Emission source’	means a separately identifiable part of an installation or a process within an installation, from which relevant greenhouse gases are emitted
EU ETS	means the system for greenhouse gas emissions allowance trading within the Union in respect of activities listed in Annex I to Directive 2003/87/EC other than aviation activities

Term	Definition
‘EU ETS allowance’	means an allowance as defined in Article 3, point (a), of Directive 2003/87/EC in respect of activities listed in Annex I to that Directive other than aviation activities
‘Fossil carbon’	means inorganic and organic carbon that is not biomass
‘Fossil fraction’	means the ratio of fossil carbon to the total carbon content of a fuel or material, expressed as a fraction
‘Fugitive emissions’	means irregular or unintended emissions from sources that are not localised, or too diverse or too small to be monitored individually
‘Functional unit’	means the reference unit used for the calculation of embedded emissions in goods
‘Goods’	means goods listed in Annex I [to the CBAM the CBAM Regulation]
‘Greenhouse gases’	mean greenhouse gases as specified in Annex I [to the CBAM the CBAM Regulation] in relation to each of the goods listed in that Annex
‘Heat meter’	means a thermal energy meter or any other device to measure and record the amount of thermal energy produced based upon flow volumes and temperatures
‘Importer’	means either the person lodging a customs declaration for release for free circulation of goods or a bill of discharge in accordance with Article 175(5) of Delegated Regulation (EU) 2015/2446 in its own name and on its own behalf or, where the customs declaration is lodged by an indirect customs representative in accordance with Article 18 of Regulation (EU) No 952/2013, the person on whose behalf such a declaration is lodged
‘Importation’	means release for free circulation as provided for in Article 201 of Regulation (EU) No 952/2013
‘Indirect customs Representative’	A customs representative appointed by an importer established in a Member State. If the importer appoints an indirect customs representative and they agree, they can act as an authorised CBAM declarant and submit the application for an authorisation on behalf of the importer. The indirect customs representative is liable for all operations made for the importers it represents
‘Indirect emissions’	means emissions from the production of electricity which is consumed during the production processes of goods, irrespective of the location of the production of the consumed electricity
‘Inherent CO₂’	means CO ₂ which is part of a source stream.
‘Installation’	means a stationary technical unit where a production process is carried out

Term	Definition
‘Legal metrological control’	means the control exercised by a public authority or regulator of the measurement tasks intended for the field of application of a measuring instrument, for reasons of public interest, public health, public safety, public order, protection of the environment, the levying of taxes and duties, the protection of consumers and fair trading
‘Measurable heat’	means a net heat flow transported through identifiable pipelines or ducts using a heat transfer medium, such as, in particular, steam, hot air, water, oil, liquid metals and salts, for which a heat meter is or could be installed
‘Measurement point’	means the emission source for which continuous emission measurement systems (CEMS) are used for emission measurement, or the cross-section of a pipeline system for which the CO ₂ flow is determined using continuous measurement systems
‘Measurement system’	means a complete set of measuring instruments and other equipment used to determine variables for the monitoring and calculation of emissions
‘Minimum requirements’	means monitoring methods using the minimum efforts allowed for determining data in order to result in emission data acceptable for the purpose of the CBAM Regulation.
‘Mixed fuel’	means a fuel which contains both biomass and fossil carbon
‘Mixed material’	means a material which contains both biomass and fossil carbon
‘Multifunctional process’	means a process that delivers multiple outputs or whose outputs feed into several production processes
‘Net calorific value’ (NCV)	means the specific amount of energy released as heat when a fuel or material undergoes complete combustion with oxygen under standard conditions, less the heat of vaporisation of any water formed
‘Non-CBAM good’	means any good produced in the installation that is not included in Annex I to the CBAM Regulation
‘Non-measurable heat’	means all heat other than measurable heat
‘Operator’	means any person who operates or controls an installation in a third country, including a parent company that controls an installation in a third country
‘Oxidation factor’	means the ratio of carbon oxidised to CO ₂ as a consequence of combustion to the total carbon contained in the fuel, expressed as a fraction, considering carbon monoxide (CO) emitted to the atmosphere as the molar equivalent amount of carbon dioxide (CO ₂)
‘Power purchase agreement’	means a contract under which a person agrees to purchase electricity directly from an electricity producer

Term	Definition
‘Precursor’	means any input material into a production process included in the list of goods set out in Annex I to the CBAM Regulation;
‘Preliminary emission factor’	means the assumed total emission factor of a fuel or material based on the carbon content of its biomass fraction and its fossil fraction before multiplying it by the fossil fraction to produce the emission factor
‘Production process’	means the chemical and physical processes carried out to produce goods in an installation
‘Production route’²⁰	means a specific technology used in a production process to produce goods
‘Process emissions’	means greenhouse gas emissions other than combustion emissions occurring as a result of intentional and unintentional reactions between substances or their transformation, for a primary purpose other than the generation of heat, including from the following processes: (a) the chemical, electrolytic or pyrometallurgical reduction of metal compounds in ores, concentrates and secondary materials; (b) the removal of impurities from metals and metal compounds; (c) the decomposition of carbonates, including those used for flue gas cleaning; (d) chemical syntheses of products and intermediate products where the carbon bearing material participates in the reaction; (e) the use of carbon containing additives or raw materials; (f) the chemical or electrolytic reduction of metalloid oxides or non-metal oxides such as silicon oxides and phosphates.
‘Proxy data’	means annual values which are empirically substantiated or derived from accepted sources and which an operator uses to substitute a data set ²¹ for the purpose of ensuring complete reporting
‘Rebate’	means any amount that reduces the amount due or paid by a person liable for the payment of a carbon price, before its payment or after, in a monetary form or in any other form.
‘Recommended improvements’	means monitoring methods which are proven means to ensure that data are more accurate or less prone to mistakes than by mere application of minimum requirements
‘Related emissions’	means: – the direct emissions caused by the source stream or emission source concerned; – emissions attributed to a quantity of measurable heat; – the indirect emissions related to the quantity of electricity concerned; – embedded emissions of a material produced or of a precursor consumed.

²⁰ Note that different production routes can fall within the same production process.

²¹ Refers to the activity data or the calculation factors.

Term	Definition
‘Reporting period’	means the period corresponding to the calendar year during which the good was produced and used by the authorised CBAM declarant as reference for the determination of embedded emissions
‘Residue’	means a substance that is not the end product that a production process directly seeks to produce; it is not a primary aim of the production process and the process has not been deliberately modified to produce it
‘Simple goods’	means goods produced in a production process requiring exclusively input materials (precursors) and fuels having zero embedded emissions
‘Source stream’	means either of the following: (a) a specific fuel type, raw material or product giving rise to emissions of relevant greenhouse gases at one or more emission sources as a result of its consumption or production; (b) a specific fuel type, raw material or product containing carbon and included in the calculation of greenhouse gas emissions using a mass-balance method
‘Specific embedded emissions’	means the embedded emissions of one tonne of goods, expressed as tonnes of CO ₂ e emissions per tonne of goods
‘Standard conditions’	means temperature of 273,15 K and pressure conditions of 101 325 Pa defining normal cubic metres (Nm ³)
‘System boundary’	means the group of chemical or physical processes included in the calculation of embedded emissions of goods under the same aggregated goods category. The system boundaries shall cover direct emissions, indirect emissions for goods not listed in Annex II to the CBAM Regulation, and the embedded emissions of any precursor.
‘Third country’	means a country or territory outside the customs territory of the [European] Union
‘Tonne of CO₂(e)’	means one metric tonne of carbon dioxide (‘CO ₂ ’), or an amount of any other greenhouse gas listed in Annex I [to the CBAM Regulation] with an equivalent global warming potential [‘CO ₂ e’]
‘Transmission system operator’	means an operator as defined in Article 2, point (35) of Directive (EU) 2019/944 of the European Parliament and of the Council (²²).

²² Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125).

Term	Definition
‘Uncertainty’	means a parameter, associated with the result of the determination of a quantity, that characterises the dispersion of the values that could reasonably be attributed to the particular quantity, including the effects of systematic as well as of random factors, expressed in percent, and describes a confidence interval around the mean value comprising 95% of inferred values taking into account any asymmetry of the distribution of values
‘Waste’	means any substance or object which the holder discards or intends or is required to discard, excluding substances that have been intentionally modified or contaminated in order to meet this definition
‘Waste gas’	means a gas containing incompletely oxidised carbon in a gaseous state under standard conditions which is a result of any of the processes listed under ‘process emissions’

Annex B

List of abbreviations

Abbreviation	Full term
AD	Activity Data
AEM	Anode Effect Minutes
AEO	Anode Effect Overvoltage
AL	Activity Level
AOD	Argon Oxygen Decarburisation.
BAT	Best Available Techniques
BF	Biomass Fraction
BFG	Blast Furnace Gas
BM	Benchmark
BOF	Basic Oxygen Furnace
BOFG	Basic Oxygen Furnace Gas
BREFs	Best Available Techniques Reference documents
CA	Competent Authority
CBAM	Carbon Border Adjustment Mechanism, the
CCR	Clinker to Cement Ratio
CCS	Carbon Capture and Storage
CCU(S)	Carbon Capture and Utilisation (and Storage)
CEMS	Continuous Emission Measurement Systems
CF	Conversion Factor
CFP	Carbon footprint of products
CHP	Combined Heat and Power
CKD	Cement Kiln Dust
CN	Combined nomenclature
COG	Coke Oven Gas
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EEA	European Economic Area (EU-27, Iceland, Norway, Liechtenstein)
EF	Emission Factor
EFTA	European Free Trade Area
EORI	Economic Operator Registration and Identification
ETS	Emissions Trading System
EU ETS	EU Emissions Trading System
EUA	EU Allowances (used in the EU ETS)
EUR	Euro (currency)

Abbreviation	Full term
FAR	Free Allocation Rules (Regulation 2019/331) ²³
FIFO	First-In-First-Out principle
GHG	Greenhouse Gas
GWP	Global Warming Potential
HBI	Hot Briquetted Iron
HS	Harmonised System (for international trade)
IEA	International Energy Agency
ISO	International Organization for Standardization
LULUCF	Land-use, land change and forestry (criteria)
MP	Monitoring Plan
MRR	Monitoring and Reporting Regulation (Regulation 2018/2066) ²⁴
MRV	Monitoring, Reporting and Verification
MS	Member State(s)
MWh	Megawatt-hour
NCV	Net Calorific Value
NPI	Nickel pig iron
OF	Oxidation Factor
PCI	Pulverised Coal Injection
PEMS	Predictive Emission Monitoring System
PFC	Perfluoro-carbon
PoS	Proofs of Sustainability
RED II	Renewable Energy Directive, re-cast
SEE	Specific embedded emissions
SEFA	Specific embedded free allocation
TARIC	Integrated Tariff of the European Union database
TJ	Terajoules
TSO	Transmission System Operator
UCC	Union Custom Code
UN/LOCODE	United Nations Code for Trade and Transport Location

²³ Free Allocation Rules (Commission Delegated Regulation (EU) 2019/331 of 19 December 2018 determining transitional Union-wide rules for harmonised free allocation of emission allowances pursuant to Article 10a of Directive 2003/87/EC of the European Parliament and of the Council)

²⁴ Monitoring and Reporting Regulation (Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council and amending Commission Regulation (EU) No 601/2012